

P1. The first matrix:

w	000	001	010	011	100	101	110	111
WG	00000	00101	01011	01110	10010	10111	11001	11100
weight	0	2	3	3	2	4	3	3

$d_{\min} = 2$. detects $d_{\min} - 1 = 1$ error; corrects $\lfloor \frac{d_{\min} - 1}{2} \rfloor = 0$ error.

The second matrix:

w	000	001	010	011	100	101	110	111
WG	000000	001101	010001	011100	100101	101000	110100	111001
weight	0	3	2	3	3	2	3	4

$d_{\min} = 2$. detects $d_{\min} - 1 = 1$ error; corrects $\lfloor \frac{d_{\min} - 1}{2} \rfloor = 0$ error.

P2 (i)	abc	000	001	010	011
	f(abc)	000000111	001001110	010010101	011011100
	abc	100	101	110	111
	f(abc)	100100011	101010100	110100001	111110000

(ii) The identity 000000000 does not exist as codeword.

(iii) If $d(u, v) = n$, then $d(f(u), f(v)) = 3n$.

Thus, the ^{possible} Hamming distances between codewords are 0, 3, 6, 9.

Hence, $d_{\min} = 3$. It detects 2 errors and corrects 1 error.

P3

coset leader	000000	001011	010101	011110	100110	101101	110011	111000
000000	000000	001011	010101	011110	100110	101101	110011	111000
000001	000001	001010	010100	011111	100111	101100	110010	111001
000010	000010	001001	010111	011100	100100	101111	110001	111010
000100	000100	001111	010001	011010	100010	101001	110111	111100
001000	001000	000011	011101	010110	101110	100101	111011	110000
010000	010000	011011	000101	001110	110110	111101	100011	101000
100000	100000	101011	110101	111110	000110	001101	010011	011000
001100	001100	000111	011001	010010	101010	100001	111111	110100

P4

(a) $d(x, y) = wt(x+y) \geq 0$

(b) $d(x, y) = wt(x-y) = 0 \Leftrightarrow wt(x-y) = 0 \Leftrightarrow x = y.$

(c) $d(x, y) = wt(x+y) = d(y, x).$

(d)
$$\begin{aligned} d(x, y) + d(y, z) &= \sum |x_i - y_i| + \sum |y_i - z_i| \\ &= \sum (|x_i - y_i| + |y_i - z_i|) \\ &\geq \sum |x_i - z_i| = d(x, z). \end{aligned}$$